Dr.-Ing. Edouard Fouché

Nationality: French, German
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CONTACT

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EXPERTISE

Machine Learning
Artificial Intelligence
Statistics, Exp. Design
Industrial & Physical AI
Algorithms, Programming
Public speaking, Teaching
Visualization, Databases

PROGRAMMING

Python (~12 years)
SQL (~10 years)
Scala/Spark (~5 years)
Java, R, Bash

SOFTWARE

Git, Linux, MacOS, Docker, Kubernetes, VS-Code + Copilot

SOFT SKILLS

Research, **Leadership**
Stakeholder Management
Agile Software Development
Cultural Awareness
Dedication, Communication

LANGUAGES

French (Native)
English, German (Fluent)

INTERESTS

Photography (> 15 years)
Hiking, Travels, Cooking,
Roller Derby, Marksmanship

Data Science & AI Leader | PhD, MBA

Experience

2023-now **Data Scientist –> Senior Key Expert, Siemens AG / Siemens Healthineers**.
- Since 02/2026 at Healthineers: AI for Strategic Procurement.
- Spend analysis, negotiation support, risk quantification, supplier segmentation.
- Tech lead for large-scale API-based forecasting product (industrial AI).
- Established MLOps, experiment pipelines (A/B tests, metrics); +300% accuracy.
- Filed 6+ invention disclosures. Keynote speaker for C-level audiences.

Tech stack: Python, PyTorch, TensorFlow, Sklearn, FastAPI, Docker, MLflow, GitLab CI/CD, Azure, AWS, Databricks, Snowflake, AI agents (Copilot Studio)

2020-2023 **Senior Researcher & Lecturer, Karlsruhe Institute of Technology (KIT)**.
- Led a research group of 4 PhD students and 10+ master students.
- Published at ACM KDD, IEEE ICDM, ECML, JMLR; incl. open-source software.
- Principal lecturer for Data Science. Best elective two years in a row.
- 6-month research stay in the group of Prof. Junya Honda, Kyoto University.

2016-2020 **Researcher & Ph.D. Candidate in Data Science, KIT**.
- 8 publications (KDD, ICDM, ECML, JMLR). Advised 10+ theses.
- 5 industry partners (Daimler, BMW, IBM). Distributed pipelines (Scala/Spark).
- 3-month research stay at the Data Mining Group, Prof. Jiawei Han, UIUC.

2015-2016 **Software Developer, IBM R&D GmbH, Böblingen.**
- Developing R and Python libraries for data science on IBM database systems.

Education

2020-2021 **MBA, Quantic School of Business and Technology.** (fka "Smartly")
- Online, 7% admission rate. Graduated with honors (top 10% of cohort).

2016-2020 **Ph.D. (Dr.-Ing.) in Data Science, KIT.**
- Thesis: Estimating Dependency, Monitoring and Knowledge Discovery in High-Dimensional Data Streams. (Advisor: Prof. Klemens Böhm)
- Summa Cum Laude (top 10%). Helmholtz Doctoral Prize. German GPA: 1.0

2015-2016 **M.Sc. Computer Science, KIT.**
- Thesis: Fast In-Database Feature Selection. (Advisor: Prof. Emmanuel Müller)
- German GPA: 1.4 (very good)

2010-2015 **Diplôme d'Ingénieur (Dipl.-Ing.), ESIEE Paris, France.**
- Major: Computer Science. Erasmus study at KIT, Germany.
- Top of the class. Final internship at IBM R&D (6 months).

Awards

2025 **Nominated Senior Key Expert in Data Science (Siemens AG)**
Key experts serve as strategic leaders and subject matter authorities.

2024 **Best Elective Lectures: Data Science 1 & Data Science 2.**
KIT Faculty of Computer Science – two times in a row.

2023 **Leadership Academy Fellow (German Scholars Organization).**
10K€ grant for leadership training, networking, mentoring.

2021 **Elite Program for Postdocs (Baden-Württemberg Stiftung).**
Research funding of 150K€ and trainings for early career researchers.

2020 **Doctoral Prize (Helmholtz Association).** 5K€ + 12K€ travel scholarship
The Helmholtz Association is the largest german scientific organisation.

2019 **Best Paper Award (SSDBM: Statistical and Scientific Database Management).**
Paper: Monte Carlo Dependency Estimation.

2017 **Software Campus Fellow (Federal Ministry of Education and Research).**
100K€ research grant, training, mentoring.

2015 **Parcours d'Excellence (ESIEE Paris).** Top 1% of the cohort.